
INTERNATIONAL GCSE BIOLOGY 9201/2

Paper 2

Mark Scheme
November 2019

Version: 1.0 Final



1 9 B Y 9 2 0 1 2 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from aqa.org.uk

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	cell → tissue → organ	1 mark for cell directly before tissue 1 mark for tissue directly before organ	2	AO1 3.1.2
01.2	A = breathing / respiratory B = circulatory C = nervous	allow phonetic allow cardiovascular	1 1 1	AO2 3.1.2d 3.2.3a 3.2.5a 3.4.1a
01.3	Organ system A B C Function allows the body to move and maintain its posture breaks down food so it can be absorbed delivers oxygen and nutrients to every part of the body enables the body to react to its surroundings exchanges gases with the environment	1 mark per correct line an additional line from an organ system cancels any mark for that organ system	3	AO1 3.1.2d 3.2.3a 3.2.5a 3.4.1a
Total			8	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.1	A		1	AO1 3.2.4a
02.2	2		1	AO2 3.1.3c
02.3	gall bladder urea		1 1	AO1 3.4.3c 3.2.4c
02.4	(more likely to be) a closer tissue match (so) less chance of rejection	allow correct description of specific antigens allow more likely to be successful or reduces need for immunosuppressant drugs	1 1	AO2 AO3 3.2.3s
02.5	any one from: • no need to wait (for donor to die) / reduces waiting time • more organs / donors available • liver in better condition	allow less chance of rejection if not given in 02.4 allow description	1	AO3 3.2.3s
02.6	they still have one kidney which is enough (for a healthy life)	allow they have 2 kidneys	1	AO3 3.2.3.s
02.7	kidney		1	AO3 3.2.3s
02.8	the longer the time from transplantation, the lower the (percentage) survival rate		1	AO3 3.2.3s
Total			10	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	A = (cell) wall		1	AO1 3.1.1b
	B= (permanent) vacuole		1	
03.2	C		1	AO2 3.1.1a
03.3	F		1	AO2 3.5.2a

03.4	Level 3: Relevant points (reasons/causes) are identified, given in detail and logically linked to form a clear account.		5-6	AO1 AO2 3.1.1d 3.1.5g,i,j 3.2.2a,b
	Level 2: Relevant points (reasons/causes) are identified, and there are attempts at logically linking. The resulting account is not fully clear.		3-4	AO1 AO2 3.1.1d 3.1.5g,i,j 3.2.2a,b
	Level 1: Points are identified and stated simply, but their relevance is not clear and there is no attempt at logical linking.		1-2	AO1 AO2 3.1.1d 3.1.5g,i,j 3.2.2a,b
	No relevant content		0	
	Indicative content • root hair / extension or cell / cytoplasm projection • for a larger surface area • to increase (rate of) absorption of water / ions • by osmosis / active transport / diffusion • active transport • requires (lots of) energy • from respiration • in (lots of) mitochondria • uptake of mineral ions requires energy / active transport • because ions are absorbed against a concentration gradient • thin cell wall reduces barrier to transport or reduces distance for uptake • vacuole with high concentration of salts for a faster rate of water uptake by osmosis			
03.5	no (available) light underground		1	AO2
	(so) roots do not photosynthesise or do not make glucose	allow so do not need chlorophyll	1	AO3 3.1.1b 3.2.1b

03.6	(for leaf cell) any one from: • uniform shape / no elongated extension • fewer mitochondria	allow converse for root hair cell allow description of shape allow includes starch grains allow <u>thicker</u> cell wall	1	AO3 3.1.1d 3.2.2b
03.7	B		1	AO2 3.1.5e
03.8	(RBC) would burst / rupture	ignore water moves in allow cells inflate / swell up	1	AO3 3.1.1a,b 3.1.5e
03.9	has no cell wall (to resist)	allow converse for leaf cell allow only has a (weak) cell membrane	1	AO2 3.1.1a,b 3.1.5e
Total			16	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	carbohydrate (in the diet) is stored as glycogen in the muscle		1	AO4 6.1 3.2.6g
	the amount of glycogen (in the muscle) depends upon the amount of carbohydrate in the diet		1	
04.2	distance of the marathon		1	AO4 6.1
	length of time the diet was eaten		1	
04.3	$900 \div 540$ 1.6666... or 1.6* 1.7	an answer of 1.7 scores 3 marks		AO2 6.3(1,4) 3.2.6g
		allow correct value from incorrect readings	1	
		allow correct significant figures from incorrect calculation	1	
04.4	more glycogen (stored in muscles)		1	AO1 AO2 3.2.6b,f,g
	(to be) converted (back) into glucose		1	
	(so) increased respiration		1	
	(and) greater energy release (available for muscle contraction / exercise)	do not allow production / making energy	1	

04.5	oxygen supply is insufficient (so) anaerobic respiration occurs so lactic acid builds up (causing pain / fatigue)	allow lactic acid is produced	1 1 1	AO1 AO2 3.2.6i,j,l
04.6	oxygen needed to convert / break down the lactic acid (so) higher breathing rate means faster intake of oxygen or (so) higher breathing rate means faster removal of carbon dioxide (and) higher heart rate means faster supply of oxygen to cells/muscles	allow to repay the oxygen debt allow oxidation of lactic acid for first two marks	1 1 1 1	AO2 AO1 AO2 3.2.6c,j,g

04.7	any one from: - oxygen debt is less - less lactic acid accumulates (athletes) more efficient at getting oxygen to muscles - description of improved supply of oxygen	eg athletes have stronger heart muscle which increases cardiac output / blood flow / stroke volume or athletes have increased vital lung capacity which takes in more oxygen per breath	1	AO3 3.2.6l
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Total			19
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	$\frac{0.086 - 0.020}{30}$ 0.0022 / 0.002 (mmol/ dm ³ / minute)	an answer of 0.0022 or 0.002 scores 2 marks allow tolerance of +/- ½ a small square allow 2.2 x 10 ⁻³ / 2 x 10 ⁻³ but do not allow 2.0 x10 ⁻³	1 1	AO2 3.4.5b 6.3(3,11)
05.2	increases from 0 to 30 (minutes) / 5 to 7.2 (mmol/dm ³) decreases from 30 to 90 (minutes)/ to 4.4 (mmol/dm ³) decrease stops/plateaus from 90 (to 120 minutes) / at 4.4 (mmol/dm ³)	correct units must be given at least once for full marks allow other correct use of data if neither point given allow 1 mark for decreases from 30 (to 120 minutes) / 7.2 (to 4.4) (mmol/dm ³)	1 1 1	AO2 3.4.5b 6.3(11)
05.3	glucose (from drink) absorbed into blood (so) insulin released / produced (by pancreas) (causing) glucose to move from the blood into cells	allow (so increased) glucose stimulates insulin secretion allow liver / muscle for cells allow (so) glucose converted to glycogen	1 1 1	AO1 AO2 3.4.5b 3.4.5a
05.4	for a (valid) comparison (as) activity / movement reduces levels of (blood) glucose or activity / movement reduces levels of (blood) glucose	allow for a fair test qualified	1 1	AO4 6.1

	(as blood glucose is) used in respiration			
05.5	blood flows from capillaries into veins in capillaries glucose moves into the cells because it is needed for respiration		1 1 1	AO2 AO3 3.2.3k
05.6	starch takes time to be broken down (into glucose) (results in) a slow / steady increase in blood glucose or prevents sudden peak / high of blood glucose	allow converse argument for glucose	1 1	AO2 AO3 3.4.5d
05.7	any five from: <u>Pros</u> - no need to extract blood or good for people who don't like sight of blood or more discreet - people more likely to check concentrations as it is not painful - records history / trends of glucose concentrations (rather than just concentration at that time) or can check effects of food / diet / exercise over time period - automatic readings so don't need to remember to keep checking - better able to control blood glucose concentrations or	must include at least one pro and one con for full marks allow converse points for finger prick test	5	AO3 AO4 3.4.5d

	better at preventing low / high blood glucose concentrations - less risk of serious side effects (or described) due to very high or very low blood glucose concentrations <u>Cons</u> - needs a new sensor fitted every 14 days / inconvenience of fitting new sensor - some people found it too painful to attach (sensor) or some people found attachment uncomfortable / itchy - could be thought of as unsightly - overload of information might be confusing			
Total				20

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	mercury is absorbed by (microscopic) algae / plants (in the river) and is passed on through the food chain	allow other references to examples in food chain	1	AO2 AO3 3.3.4b 3.3.1
	organisms at each trophic level feed on lots of organisms at the previous trophic level	allow large fish eat lots of small fish	1	
	(this means) mercury concentration builds up / increases as it is not removed or mercury concentration builds up because it remains attached to proteins (in the organisms tissues)	allow bioaccumulation	1	
06.2	any three from: - correlation does not infer causation - numbers counted are quite low (and this may increase unreliability) - level of mercury pollution is not measured / results are not quantitative - salmon numbers may change due to other pollutants - there may be fluctuation in pollution during the year - salmon numbers may change due to biological factors (only) monitored every five years so do not know pollution levels between these times.	allow fish numbers could be miscounted allow described eg natural predator / prey cycles, less disease or more food available	3	AO4 6.1 3.3.4a,b

06.3	any four from: may be correct as - mean mercury concentration is 3.8 times more when fish were eaten from the polluted river or mean mercury concentration is higher by 2.53ppm when fish were eaten from the polluted river - the highest levels were in people who ate fish from a polluted river or levels between 3.2 and 17.8ppm / levels up to 17.8ppm were in people who ate fish from the polluted river may not be correct as - - mean mass of fish eaten per month was 3.8 times more from the polluted river - large variation in amount of fish eaten by both groups (higher mean mercury concentrations) might be due to other factors / named eg - lowest mercury concentration from eating fish from polluted river is smaller than lowest concentration or mean concentration from eating fish from unpolluted water or data is overlapping - no indication that person who eats the most fish has the highest mercury concentration	must include at least one correct and one incorrect for four marks accept other comparative figures accept other comparative figures	4	AO3 AO4 3.3.4b 6.1
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06.4	any three from: • smaller brain / cerebrum • shrinkage of cerebral cortex or folding of the surface of the brain is not as deep / reduction in surface area of cerebral cortex • larger ventricles • the 2 halves of the brain become more separate / pulled apart at the top	allow death / reduction in neurones	3	AO3 3.4.1 3.3.4b
06.5	mercury prevents calcium (ions) from entering neurone / synapse (so) chemical is not released from (sensory) neurone at synapse / gap (so) no (electrical) impulse along relay / motor neurone (so) effector / (eye) muscle does not respond / contract (so vision is blurred)	allow neurotransmitter for chemical	1 1 1 1	AO2 AO3 3.4.1d 3.3.4b
Total			17	